GAME BASED APTITUDE

60 PYQ-Style Questions • Practice Set 1

Placement & Competitive Exam Practice Series

Included	60 PYQ-Style Questions • Placement-focused
Useful For	TCS NQT • Cognizant • Infosys • Wipro • Accenture • Campus Placements
Topics Covered	Grid Challenge • Digit Challenge • Motion Challenge • Geo-Symmetry • Switch Challenge • Inductive Patterns
Format	10 MCQs + Answer Key per Page • Practice Set 1

These are PYQ-style practice questions modeled on common campus recruitment exam patterns.
They are designed for comprehensive practice, mental speed, and spatial logic assessment.

GAME BASED APTITUDE • GRID CHALLENGE & VISUAL MEMORY

Practice Set 1 • Section 1 of 6 • 10 Questions with Answer Key

Q1. In a 4x4 grid of flashing tiles, 5 tiles flash blue for 3 seconds. To replicate the sequence, which cognitive strategy minimizes memory decay?

A) Chunking into spatial coordinates
B) Memorizing color shades
C) Alphabetical conversion
D) Random guessing

Q2. A grid rotates 90 degrees clockwise after each round. If a dot is at position (1, 3) in a 4x4 matrix, where is it after one 90 deg clockwise rotation?

A) (3, 4)
B) (3, 1)
C) (4, 2)
D) (2, 4)

Q3. In a dual-task N-back grid game, a square lights up at row 2, col 3. If N=2, what must the player recall?

A) Position 2 steps prior
B) Position 1 step prior
C) Future predicted position
D) Only the color

Q4. A 3x3 matrix has numbers in each cell summing to 15 in every row and column (magic square). If the center cell is 5, what is opposite cell to 1?

A) 9
B) 7
C) 8
D) 6

Q5. In a memory matrix round, a pattern of 6 tiles appears and inverts colors. If the grid has 16 cells, how many cells remained unchanged?

A) 10
B) 6
C) 8
D) 12

Q6. A path moves on a 5x5 grid: 2 steps Right, 3 steps Down, 1 step Left, 2 steps Up. Net displacement from start (0,0) is:

A) (1, 1)
B) (1, -1)
C) (2, -1)
D) (0, 1)

Q7. Which visual property is processed fastest by pre-attentive sensory memory in grid challenges?

A) Color contrast / Pop-out effect
B) Number parity
C) Font family
D) Semantic meaning

Q8. In an 8x8 chessboard grid challenge, what is the minimum knight moves required to travel from a1 to b3?

A) 1 move
B) 2 moves
C) 3 moves
D) 4 moves

Q9. In a matrix dot-connection game, how many straight lines are needed to join all points in a 3x3 grid without lifting the pen?

A) 4 lines
B) 5 lines
C) 6 lines
D) 3 lines

Q10. A symmetric grid has line symmetry across the main diagonal. If cell (2, 4) is shaded, which cell must also be shaded?

A) (4, 2)
B) (2, 2)
C) (4, 4)
D) (3, 3)

ANSWER KEY & SOLUTIONS — GRID CHALLENGE & VISUAL MEMORY (SET 1)

Q1: A • Q2: A • Q3: A • Q4: A • Q5: A

Q6: B • Q7: A • Q8: A • Q9: A • Q10: A

GAME BASED APTITUDE • DIGIT CHALLENGE & MENTAL ARITHMETIC

Practice Set 1 • Section 2 of 6 • 10 Questions with Answer Key

Q1. Using digits [2, 4, 6] exactly once each with any basic arithmetic operators (+, -, *, /), which target CANNOT be made?

A) 24
B) 48
C) 14
D) 100

Q2. In a countdown math round, target is 45. Available numbers: [3, 5, 9]. Fastest single multiplication is:

A) 5 * 9 = 45
B) (9 - 3) * 5
C) 3 * 5 * 3
D) 9 * 3 + 5

Q3. What is the missing digit in the arithmetic wheel: 4 -> 8 -> 16 -> 32 -> ?

A) 64
B) 48
C) 52
D) 60

Q4. In a speed-sum game: 17 + 28 + 33 + 42 = ?

A) 120
B) 115
C) 125
D) 130

Q5. If A + B = 14 and A * B = 48, what is the absolute difference |A - B|?

A) 2
B) 4
C) 6
D) 8

Q6. A rapid digit string displays: 4, 9, 16, 25, 36. What is the 6th number in this sequence?

A) 49
B) 42
C) 64
D) 56

Q7. In a balance beam math puzzle, 3 circles weigh equal to 2 squares. If 1 square is 6kg, weight of 1 circle is:

A) 4 kg
B) 3 kg
C) 9 kg
D) 2 kg

Q8. What is the remainder when (13 * 17 * 19) is divided by 2?

A) 1
B) 0
C) -1
D) 2

Q9. In a quick target game, you need to reach 100 starting from 0 using only +7 and -2. Minimum operations to reach 12:

A) 4 operations (+7, +7, -2)
B) 6 operations
C) 3 operations (+7, +7, -2)
D) 2 operations

Q10. If 5 # 3 = 16 and 7 # 2 = 45, what is the formula rule for x # y?

A) x^2 - y^2
B) (x + y) * 2
C) x * y + 1
D) 2x + 3y

ANSWER KEY & SOLUTIONS — DIGIT CHALLENGE & MENTAL ARITHMETIC (SET 1)

Q1: D • Q2: A • Q3: A • Q4: A • Q5: A

Q6: A • Q7: A • Q8: A • Q9: C • Q10: A

GAME BASED APTITUDE • MOTION CHALLENGE & TRAJECTORY PHYSICS

Practice Set 1 • Section 3 of 6 • 10 Questions with Answer Key

Q1. A ball travels inside a bounded polygon reflecting with angle of incidence = angle of reflection. When striking a flat horizontal floor at 45 deg, it reflects at:

A) 45 deg to normal
B) 90 deg to normal
C) 0 deg
D) 60 deg

Q2. Gear A has 20 teeth and rotates clockwise at 60 RPM. It drives Gear B with 40 teeth. Gear B rotates:

A) Counter-clockwise at 30 RPM
B) Clockwise at 30 RPM
C) Counter-clockwise at 120 RPM
D) Clockwise at 120 RPM

Q3. In an evasion game, Object X moves at 5 m/s due East while Object Y moves at 5 m/s due North. Relative speed is:

A) 5 * sqrt(2) m/s
B) 10 m/s
C) 0 m/s
D) 5 m/s

Q4. A projectile trajectory in a 2D gravity simulation has a parabolic arc. At the peak of its trajectory, its vertical velocity is:

A) 0 m/s
B) Maximum
C) Equal to horizontal velocity
D) 9.8 m/s

Q5. In a collision simulator, two equal masses traveling in opposite directions collide elastically. After collision they:

A) Swap velocities
B) Stick together
C) Stop instantly
D) Explode

Q6. A car accelerates uniformly from 0 to 20 m/s in 4 seconds. Total distance covered during acceleration is:

A) 40 meters
B) 80 meters
C) 50 meters
D) 20 meters

Q7. In a pendulum timing game, the period of a simple pendulum depends primarily on:

A) Length of string
B) Mass of bob
C) Color of bob
D) Initial push force

Q8. Three gears connected in a linear train (A - B - C). If Gear A turns clockwise, what direction does Gear C turn?

A) Clockwise
B) Counter-clockwise
C) Alternating
D) Stationary

Q9. In a maze navigation speed run, turning 180 degrees left then 90 degrees right results in a net turn of:

A) 90 degrees left
B) 90 degrees right
C) 270 degrees right
D) 0 degrees

Q10. A conveyor belt travels at 3 km/h. A robot walks forward on it at 4 km/h. To a stationary observer, the robot speed is:

A) 7 km/h
B) 1 km/h
C) 12 km/h
D) 5 km/h

ANSWER KEY & SOLUTIONS — MOTION CHALLENGE & TRAJECTORY PHYSICS (SET 1)

Q1: A • Q2: A • Q3: A • Q4: A • Q5: A

Q6: A • Q7: A • Q8: A • Q9: A • Q10: A

GAME BASED APTITUDE • GEO-SYMMETRY & SPATIAL FOLD REASONING

Practice Set 1 • Section 4 of 6 • 10 Questions with Answer Key

Q1. A square paper is folded in half diagonally, then a circular hole is punched in the center. When unfolded, how many holes appear?

A) 2 holes
B) 1 hole
C) 4 holes
D) 8 holes

Q2. Which 3D shape is formed by folding a net consisting of 6 identical connected squares?

A) Cube
B) Tetrahedron
C) Octahedron
D) Cylinder

Q3. How many axes of reflective symmetry does a regular hexagon possess?

A) 6
B) 4
C) 8
D) 12

Q4. If a 3D block has dimensions 2x3x4, how many unit cubes (1x1x1) make up its total volume?

A) 24
B) 18
C) 12
D) 20

Q5. A transparent sheet with an arrow pointing UP is flipped horizontally over its vertical axis. The arrow points:

A) Still UP
B) DOWN
C) LEFT
D) RIGHT

Q6. Which shape possesses rotational symmetry of order 4 but no lines of reflective symmetry?

A) Pinwheel with 4 identical curved blades
B) Square
C) Rectangle
D) Circle

Q7. When looking at an isometric projection of stacked cubes, hidden cubes must be inferred by:

A) Height support levels
B) Lighting reflections
C) Random probability
D) Perspective angle

Q8. A cube has faces colored: Top/Bottom Red, Left/Right Green, Front/Back Blue. How many faces are Green?

A) 2
B) 4
C) 1
D) 3

Q9. Unfolding a triangular prism net yields how many rectangular faces?

A) 3 rectangular faces
B) 2 rectangular faces
C) 4 rectangular faces
D) 1 rectangular face

Q10. A tessellation pattern covers a 2D plane with repeating shapes. Which regular polygon CANNOT tessellate a plane alone?

A) Regular Pentagon
B) Regular Hexagon
C) Equilateral Triangle
D) Square

ANSWER KEY & SOLUTIONS — GEO-SYMMETRY & SPATIAL FOLD REASONING (SET 1)

Q1: A • Q2: A • Q3: A • Q4: A • Q5: A

Q6: A • Q7: A • Q8: A • Q9: A • Q10: A

GAME BASED APTITUDE • SWITCH CHALLENGE & LOGIC GATE NETWORKS

Practice Set 1 • Section 5 of 6 • 10 Questions with Answer Key

Q1. In a logic circuit puzzle, inputs A=1 and B=0 pass through an XOR gate. The output is:

A) 1
B) 0
C) High Impedance
D) Undefined

Q2. If two single-pole single-throw switches are connected in SERIES, the light bulb turns ON when:

A) Both switches are closed
B) Either switch is closed
C) Both switches are open
D) One switch is open

Q3. A NAND gate produces an output of 0 only when inputs are:

A) Both 1
B) Both 0
C) 0 and 1
D) Floating

Q4. Three toggle switches control 3 lamps. If each toggle inverts state (0->1, 1->0), what state is achieved after 5 toggles from 0?

A) 1 (ON)
B) 0 (OFF)
C) Tri-state
D) Disabled

Q5. Which combination of logic gates can implement any arbitrary Boolean function (universal gate)?

A) NAND or NOR
B) AND and OR only
C) XOR only
D) NOT only

Q6. In a truth table challenge: Output is 1 if and only if exactly one input is 1. This gate is:

A) 2-input XOR gate
B) AND gate
C) OR gate
D) NOR gate

Q7. If an input signal passes through two consecutive NOT (inverter) gates, the output is:

A) Same as original input
B) Inverted input
C) Always 0
D) Always 1

Q8. In a 4-switch combination lock, each switch has 2 states (Up/Down). How many total possible combinations exist?

A) 16
B) 8
C) 12
D) 32

Q9. Two parallel switches connected to a DC circuit represent which Boolean logic function?

A) OR function
B) AND function
C) NOT function
D) XNOR function

Q10. What is the output of (A AND B) OR (A AND (NOT B)) simplified by Boolean algebra laws?

A) A
B) B
C) NOT A
D) 1

ANSWER KEY & SOLUTIONS — SWITCH CHALLENGE & LOGIC GATE NETWORKS (SET 1)

Q1: A • Q2: A • Q3: A • Q4: A • Q5: A

Q6: A • Q7: A • Q8: A • Q9: A • Q10: A

GAME BASED APTITUDE • INDUCTIVE REASONING & SEQUENCE PATTERNS

Practice Set 1 • Section 6 of 6 • 10 Questions with Answer Key

Q1. Find the next item in pattern: Circle (1 dot) -> Square (2 dots) -> Pentagon (3 dots) -> ?

A) Hexagon (4 dots)
B) Heptagon (5 dots)
C) Triangle (4 dots)
D) Circle (5 dots)

Q2. In a tile shifting sequence, a black square shifts 1 position clockwise each frame. After 4 frames in a 4-corner box:

A) Returns to original corner
B) In opposite corner
C) In center
D) Disappears

Q3. Series: 2, 6, 12, 20, 30, ?. What is the next number?

A) 42
B) 40
C) 36
D) 48

Q4. In a matrix of symbols, each row contains [Star, Diamond, Triangle]. If Row 3 has Diamond and Star, what is missing?

A) Triangle
B) Square
C) Circle
D) Heart

Q5. A rule states: Number of sides increases by 1, color alternates (Black -> White). After Black Hexagon comes:

A) White Heptagon (7 sides)
B) Black Heptagon
C) White Octagon
D) Black Pentagon

Q6. Which reasoning style generalizes from specific observed instances to broad probabilistic rules?

A) Inductive Reasoning
B) Deductive Reasoning
C) Reductio ad absurdum
D) Circular reasoning

Q7. Series of letters: Z, X, V, T, R, ?. The next letter is:

A) P
B) Q
C) N
D) O

Q8. In an odd-one-out shape game: Triangle, Square, Pentagon, Hexagon, Circle. Which is anomalous?

A) Circle (no straight edges/vertices)
B) Triangle
C) Square
D) Hexagon

Q9. Sequence of clock faces: 12:00, 1:15, 2:30, 3:45, ?. Next time is:

A) 5:00
B) 4:50
C) 5:15
D) 4:30

Q10. A number grid pattern: Row 1 [2, 3, 5], Row 2 [7, 11, 13], Row 3 [17, 19, ?]. What is the missing prime number?

A) 23
B) 21
C) 27
D) 29

ANSWER KEY & SOLUTIONS — INDUCTIVE REASONING & SEQUENCE PATTERNS (SET 1)

Q1: A • Q2: A • Q3: A • Q4: A • Q5: A

Q6: A • Q7: A • Q8: A • Q9: A • Q10: A